# CHEMISTRY

1. Which of the following species are paramagnetic:

$O_2, N_2, F_2, Cl_2$

(1) $N_2$, $F_2$

(2) Only $O_2$

(3) Only $Cl_2$ & $N_2$

(4) $O_2, Cl_2$

Ans. (2)

Sol. $O_2$ : Paramagnetic

$N_2$ : Diamagnetic

$F_2$ : Diamagnetic

$Cl_2$ : Diamagnetic

2. Which of the following is the ratio of 5$^{th}$ Bohr orbit ($r_5$) of $He^+$ & $Li^{2+}$?

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) $\frac{9}{4}$

(4) $\frac{4}{9}$

Ans. (2)

Sol. $r_5\left(He^+\right) = a_0 \times \frac{5^2}{2}$

$r_5\left(Li^{+2}\right) = a_0 \times \frac{5^2}{3}$

$$r_5\left(Li^{+2}\right) = \frac{5^2}{3}$$

$$\frac{r_5\left(He^{+}\right)}{r_5\left(Li^{+2}\right)} = \frac{3}{2}$$

3. Which of the following pair of ions have equal number of upaired electrons

(1) $V^{2+}$ and $Ni^{2+}$

(2) $Cr^{2+}$ and $Mn^{2+}$

(3) $Fe^{2+}$ and $Sc^{2+}$

(4) $Mn^{3+}$ and $Fe^{2+}$

Ans. (4)

Sol. $Mn^{3+} \rightarrow d^4$; 4 unpaired electron

$Fe^{+2} \rightarrow d^6$; 4 unpaired electron

4. Which of the following is correct order of size?

(1) B > Al > Ga > In > Tl

(2) B < Ga < Al < In < Tl

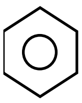
(3) B < Al < Ga < In < Tl

(4) B > Al < Ga < In < Tl

Ans. (2)

Sol. Correct order of size : B < Ga < Al < In < Tl

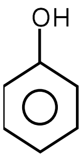
5. In the reaction sequence :

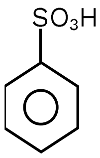
 + Oleum $\longrightarrow$ X

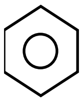
$X \xrightarrow[\text{(ii)}H^+, \Delta]{\text{(i) NaOH}} Y$

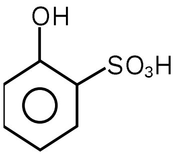
$Y \xrightarrow{\text{Zn dust}} Z$

The compound Z is :

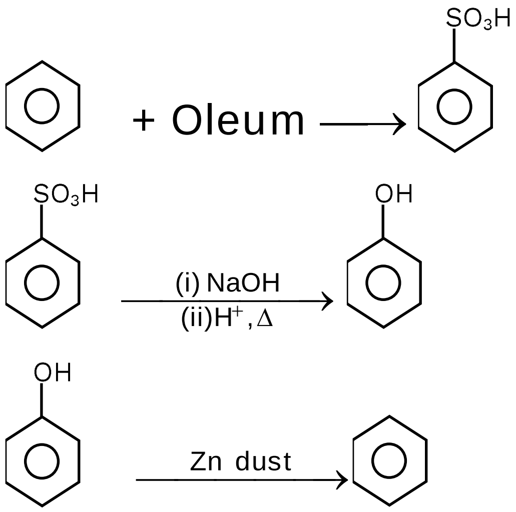
(1) 


(2) 


(3) 

(4) 


Ans. (3)

Sol. 


6. If rate = $K[A]^m [B]^n$ and concentration of A is doubled and B is halved, then

(1) Rate becomes $2^{m-n}$ times

(2) Rate becomes $2^{n-m}$ times

(3) Rate becomes 2 times

(4) Rate becomes $2^m$ times

Ans. (1)

Sol. rate = $K[A]^m [B]^n$

New rate (r') = $K[2A]^m \left[\frac{B}{2}\right]^n$

$= 2^{m-n} K[A]^m [B]^n$

$= 2^{m-n} \times r$

7. One mole of an ideal gas expands from $10 dm^3$ to $20 dm^3$ through isothermal reversible process. Find $\Delta U$, q & w

(1) $\Delta U = 0$, $q = 0$, $w = 0$

(2) $\Delta U = 0$, $q \neq 0$, $w \neq 0$

(3) $\Delta U \neq 0$, $q = 0$, $w \neq 0$

(4) $\Delta U = 0$, $q \neq 0$, $w = 0$

Ans. (2)

Sol. For isothermal reversible process

T = Constant

So $\Delta U = 0$

FLOT $\Delta U = q + w$

$q = -w$

8. Number of Geometrical isomers in $[Ma_3b_3]$ and $[Ma_2(AA)_2]$ are :

(1) 2, 3

(2) 2, 2

(3) 3, 2

(4) 2, 4

Ans. (3)

Sol. $[CrCl_3(Py)_3]$
$[Ma_3b_3]$
Two Isomers
Fac & mer

$[CrCl_2(C_2O_4)_2]$
$M(AA)_2\ (a_2)$
Shows Cis & trans
Cis $\rightarrow$ (optically active)
trans$\rightarrow$optically Inactive
Total = 3 Isomer

9. Which of the following is correct option regarding 1s orbital
(1) It is symmetrical
(2) It is non-symmetrical
(3) It is directional
(4) It has two radial nodes

Ans. (1)

Sol. Shape of s-orbital is spherical
for radial node
n-l-1
for 1s
l=0 & n=1
so radial node $\Rightarrow$ 1-0-1 = 0

10. Statement-I: $CH_3-CH=CH-CH=O$ is greater dipole moment than $CH_3-CH_2-CH_2-CH=O$

Statement-II: In $CH_3-CH_2\overset{x}{—}CH_2-CH=O$ and $CH_3-CH\overset{y}{=}CH-CH=O$, bond length is y > x.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are false

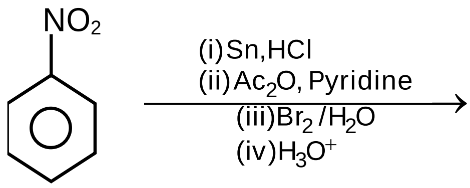
(2) Statement-I is false but Statement-II is true

(3) Both Statement I and Statement-II are true

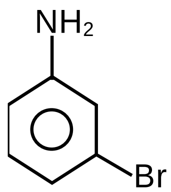
(4) Statement-I is true but Statement-II is false

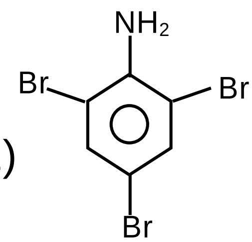
Ans. (4)

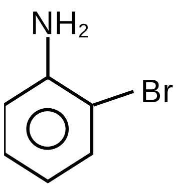
Sol.

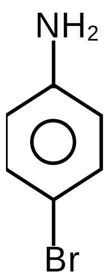
11. 

Major product is

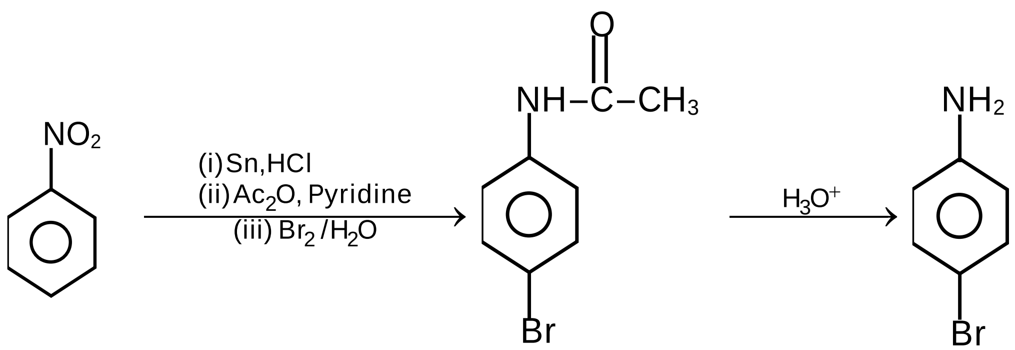
(1) 


(2) 

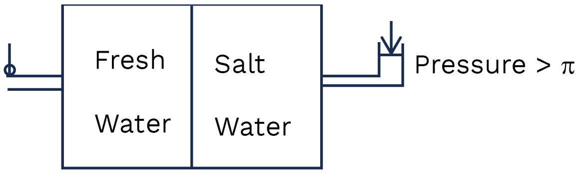

(3) 


(4) 


Ans. (4)

Sol. 

12. Observe the following diagram.




For reverse osmosis, which of the following can be used for porous membrane?

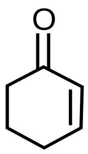
(1) Cellulose acetate (2) Porous silicate

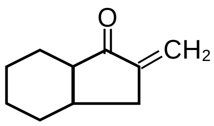
(3) Silicone (D) Glass membrane

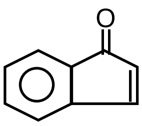
Ans. (1)

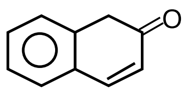
Sol. NCERT

13. Which product is not formed by intramolecular aldol condensation?

(1) 

(2) 

(3) 

(4) 

Ans. (2)

Sol. -

14. Dipeptide Gly-Ala is formed by

(1) $H_2N-CH_2-\underset{\underset{O}{\|}}{C}-Cl$, $H_2N-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{O}{\|}}{C}-OH$

(2) $H_2N-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{O}{\|}}{C}-Cl$, $H_2N-CH_2-\underset{\underset{O}{\|}}{C}-OH$

(3) $H_2N-CH(CH_3)-C(=O)-Cl$, $H_3\overset{\oplus}{N}-CH_2-C(=O)-OH$

(4) $H_2N-CH(CH_3)-C(=O)-Cl$, $H_2N-CH(CH_3)-C(=O)-OH$

Ans. (1)

Sol. -

15. In lead storage battery during charging oxidation state of lead show changes at anode from $x_1$ to $y_1$. at cathode from $x_2$ to $y_2$ Find valuer of $x_1$,$y_1$,$x_2$,$y_2$.

(1) $x_1 = +2, y_1 = 0$
$x_2 = +2, y_2 = +4$

(2) $x_1 = +4, y_1 = 0$
$x_2 = +2, y_2 = +4$

(3) $x_1 = 0, y_1 = +2$
$x_2 = +4, y_2 = +2$

(4) $x_1 = +2, y_1 = 0$
$x_2 = +4, y_2 = 0$

Ans. (1)

Sol. At anode $\overset{+2}{Pb}SO_4 \longrightarrow \overset{0}{Pb}$

At cathode $\overset{+2}{Pb}SO_4 \longrightarrow \overset{+4}{Pb}O_2$

16. In wheat Fe is present in the form of $FeSO_4.7H_2O$. If concentration of iron is 12ppm. Calculate mass of $FeSO_4.7H_2O$ is present in 150 Kg of wheat.

Ans. 8

Sol. mass of Fe in $10^3$kg of wheat = 12g

$\therefore$ mass of Fe in 150kg of what = $\frac{12}{10^3} \times 150$

$= 12 \times 0.15\text{gm}$

$= 1.8\text{g}$

mole of Fe = $\frac{1.8}{56}$

mole of $FeSO_4.7H_2O$ = $\frac{1.8}{56}$

$\therefore$ mass of $FeSO_4.7H_2O$ = $\frac{1.8}{56} \times 246 = 7.9\text{gm}$

17. 0.01 M HX ($K_a = 4 \times 10^{-10}$) is dilluated till the solution has pH = 6. If the new concentration is $x \times 10^{-4}$ M then find x.

Ans. (25)

Sol. Given pH = 6

$$\left[H^+\right] = 10^{-6} = \sqrt{k_a.C}$$

$$10^{-12} = \left(4 \times 10^{-10}\right)(C)$$

$$C = \frac{1}{4} \times 10^{-2} = 0.25 \times 10^{-2}$$

$$= 25 \times 10^{-4}$$

18. $C_3H_6O(X) \xrightarrow{\text{Reduction}} Y \xrightarrow{\text{HBr}} Z \xrightarrow{\text{Mg/D.E.}} A$

$\xrightarrow[\text{(ii)}H_2O]{\text{(i)}X}$ 2, 3-dimethyl butan-2-ol

X,Y and Z are

Ans.

Sol.

19. For a chemical reaction, $\Delta H$ and $\Delta S$ are positive. If equilibrium temperature is $T_e$. Temperature (T) at which reaction is spontaneous is

(1) $T < T_e$

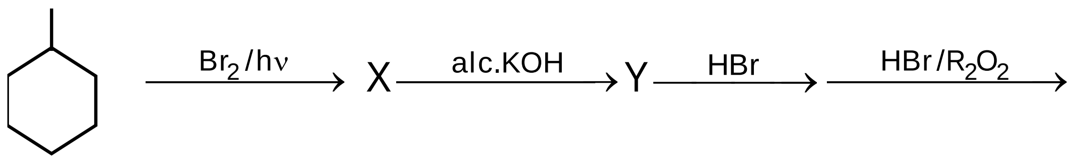
(2) $T = T_e$

(3) $T > T_e$

(4) $T = \frac{T_e}{5}$

Ans. (3)

Sol. -

20. 


Major product is:

(1)

(2)

(3)

(4)

Ans. -

Sol. -

21. In which of the following complex CFSE = 0 and $\mu$ = 5.97 BM.

(1) $[Fe(SCN)_6]^{4-}$

(2) $[Mn(H_2O)_6]^{2+}$

(3) $[Fe(CN)_6]^{4-}$

(4) $[Co(F)_6]^{3-}$

Ans. (2)

Sol. -

22. For an elementary reaction.

$A_2 + B_2 \rightleftharpoons 2AB$

$(E_a)_{forward}$ = 180 kJ/mol

$(E_a)_{backward}$ = 200 kJ/mol

A catalyst is added which reduces activation energy by 100 kJ.

Which of the following is correct.

(1) $\Delta H_{reaction}$ = +20 KJ/mol

(2) A non spontaneous becomes spontaneous on adding catalyst.

(3) $\Delta G_{reaction}$ remains unaffected on catalyst.

(4) $\Delta H$ reaction depends on the presence of catalyst.

Ans. (3)

Sol.